

HAT-P-36b

R-0798

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_250408 · created 2026-07-28T08:56:21+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460773.6965 +/- 0.0062 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.073 +/- 0.03
Transit depth (Rp/Rs)^2	0.53 +/- 0.44 [%]
Orbital Inclination (inc)	80.5 +/- 2.9
Airmass coefficient 1 (a1)	1.059 +/- 0.017
Airmass coefficient 2 (a2)	-0.054 +/- 0.027
Scatter in the residuals of the lightcurve fit is	1.62 %
Best Comparison Star	#1 - [113, 143]
Optimal Aperture	3.05
Optimal Annulus	6.72
Transit Duration (day)	0.062 +/- 0.021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.073 ± 0.03 vs 0.126 (deviation 1.34σ)
Duration fit vs published	0.062 d vs 0.0925 d (deviation 1.1σ)
Mid-time O–C	8.1 min
Depth SNR	1.8
Residual scatter	1.62 %

Light curve

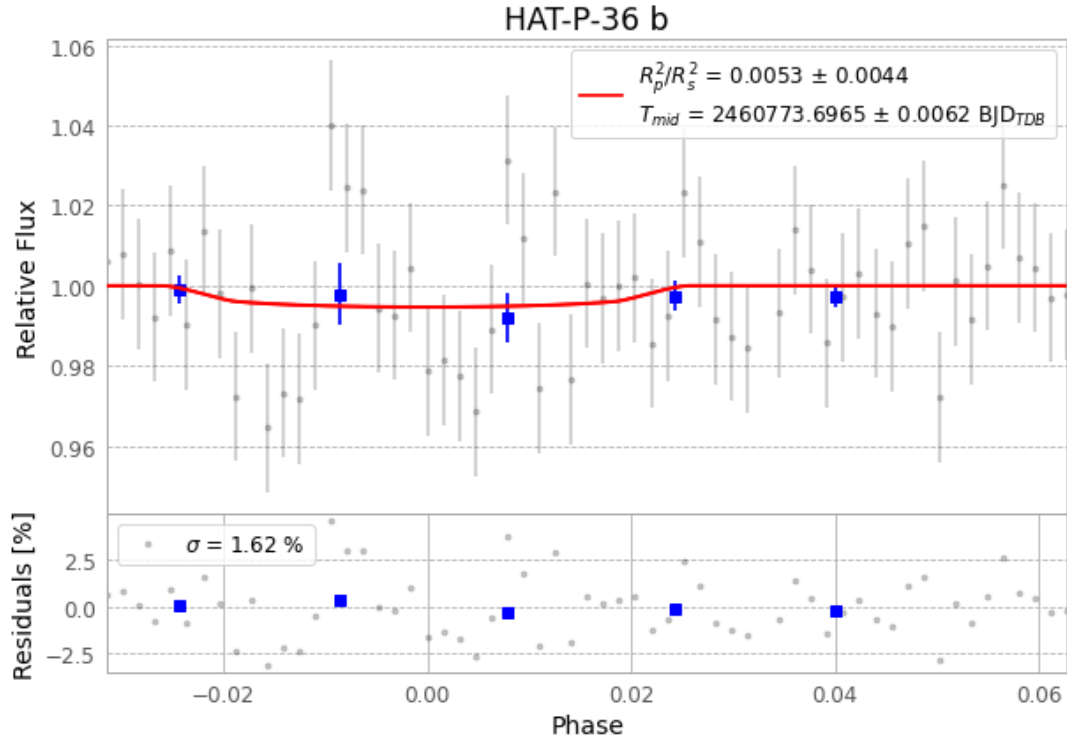


Figure 1 — FinalLightCurve_HAT-P-36 b_2025-04-07.png (SHA-256 4ea537deb95072cb...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [385, 210], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3ac1bfe7e2f7c0d4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

61 FITS frames (40 MB), HATP-36250408033615.FITS → HATP-36250408063615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-250408.log (cdaa0d5c3abe...), FinalLightCurve_HAT-P-36 b_2025-04-07.png (4ea537deb950...), FinalLightCurve_HAT-P-36 b_2025-04-07.csv (c53d7f3e9bbb...)

Compute resources

Wall time 130.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 08:56Z.